

ASSESSMENT TOOL 2 – Technical Presentation

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
5	22	19	20	15	10	86

Faculty In-charge	Course Coordinator	Program Director
Dr. T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



NLP COURSE • TECHNICAL PRESENTATION

The History of Natural Language Processing

From symbolic rule-based systems to large language models — six decades of teaching machines to understand human language.

Presented in NLP Coursework

Introduction

What is Natural Language Processing?

Natural Language Processing (NLP) is the field of artificial intelligence focused on enabling computers to understand, interpret, and generate human language.

Its history reflects the broader story of AI itself — moving from hand-written rules, to statistical models learned from data, to today's deep neural networks trained on massive text corpora.

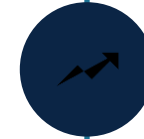
Why study its history?

Every major idea in modern NLP — embeddings, attention, transfer learning — builds directly on breakthroughs (and dead ends) from earlier eras.



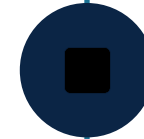
1950s

Symbolic Rules



1990s

Statistical Methods



2010s

Neural Networks



2020s

Large Language Models

Early Foundations

The Symbolic & Rule-Based Era · 1950 – 1980



1950

The Turing Test

Alan Turing proposes a test of machine intelligence based on natural conversation, framing language as the benchmark for "thinking machines."



1954

Georgetown-IBM Experiment

The first public machine translation demo converts Russian sentences to English, sparking major government funding for MT research.



1957

Syntactic Structures

Noam Chomsky's formal grammar theory gives NLP its first rigorous mathematical foundation for sentence structure.



1966

ELIZA Chatbot

Joseph Weizenbaum's ELIZA simulates a psychotherapist using pattern matching — an early, influential rule-based conversational agent.

The Shift to Data-Driven NLP

Statistical Revolution → Deep Learning · 1990 – Present



1990s – 2000s

Statistical NLP

- Rules replaced by probabilities learned from large text corpora
- N-gram language models predict word sequences
- Hidden Markov Models power early speech recognition & POS tagging
- IBM's statistical machine translation models emerge



2010s – Present

Neural & Deep Learning NLP

- Word2Vec (2013) learns dense word embeddings from context
- RNNs & LSTMs model sequences, enabling early neural translation
- The Transformer (2017) introduces self-attention, replacing recurrence
- BERT, GPT, and modern LLMs achieve human-level fluency at scale

NLP in the Real World

Applications built on decades of research



Machine Translation

Google Translate & DeepL translate 100+ languages in real time.



Conversational AI

Chatbots & assistants like Siri, Alexa, and modern LLM assistants.



Search Engines

Understanding query intent to rank and retrieve relevant results.



Spam & Content Filtering

Classifying emails and moderating text content automatically.



Sentiment Analysis

Gauging public opinion from reviews and social media at scale.



Speech Recognition

Converting spoken language to text for dictation and voice control.

From ELIZA's scripted replies to today's LLMs — NLP's history is a story of scaling from rules, to statistics, to learned representations.

GEOTAG PHOTO

